

SW6100/SW6100P Memory Guide - Linux Android Wear

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Contents

1	Memory overview	3
1.1	DRAM capabilities of SW6100/SW6100P	3
1.2	Enable DRAM	4
2	Customize and validate the DRAM components	5
3	Install and test the QMVS DDR validation tool	7
3.1	Download QMVS	8
3.2	Install QMVS	8
3.3	Run QMVS validation tests	9
4	Troubleshoot DRAM issues	11
4.1	DRAM initialization	11
4.2	DRAM usage	11
5	References	13
5.1	Related documents	13
5.2	Acronyms and terms	13

1 Memory overview

Learn how low-power double data rate (LPDDR) dynamic random-access memory (DRAM) supports the SW6100/SW6100P platform, including key capabilities, memory architecture, and the DRAM initialization process during boot.

Qualcomm chip products use LPDDR DRAM to support system operations. DRAM is a type of volatile memory that stores data frequently accessed by the processor and requires constant refreshing to retain information.

The Qualcomm memory controller supports [JEDEC](#) - compliant LPDDR5 and LPDDR5X DRAM components. You can select the type of DRAM based on your device requirements.

1.1 DRAM capabilities of SW6100/SW6100P

The memory controller supports high-speed LPDDR5/LPDDR5X SDRAM in single-channel package-on-package (PoP) configurations.

- LPDDR5/LPDDR5X SDRAM supports a maximum memory density of 4 GB.
- LPDDR5/LPDDR5X SDRAM operates at a clock speed of up to 4200 MHz (1 × 16-bit).

The following figure shows the DRAM memory channel architecture for SW6100/SW6100P:

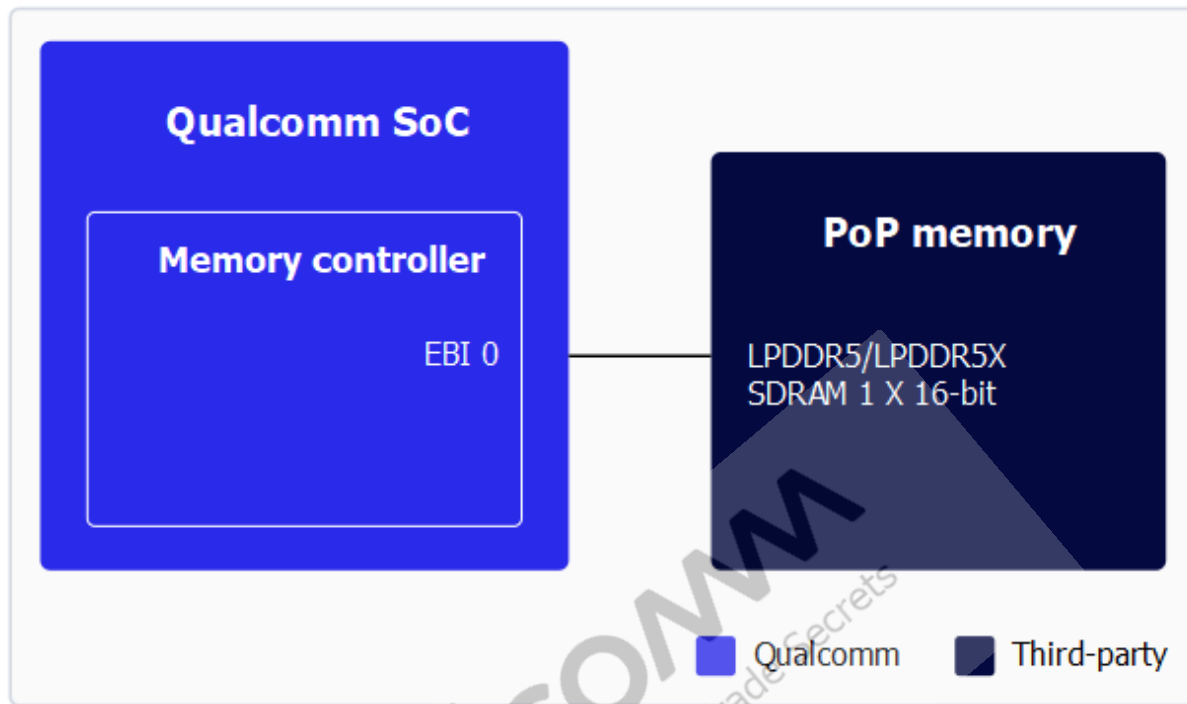


Figure : SW6100/SW6100P DRAM memory channel

1.2 Enable DRAM

The DDR system firmware (DSF) is embedded software that enables DRAM operation. The eXtensible Boot Loader (XBL) image includes the DSF, which initializes the memory subsystem and runs during the boot sequence. The software works with all the DRAM components listed in the Qualcomm preferred vendor list (PVL). For information about using custom DRAM options from the PVL, see [Customize and validate DRAM components](#).

The software doesn't require any additional modifications or configuration for the PVL DRAM components because it automatically programs all required settings and timings during the boot process.

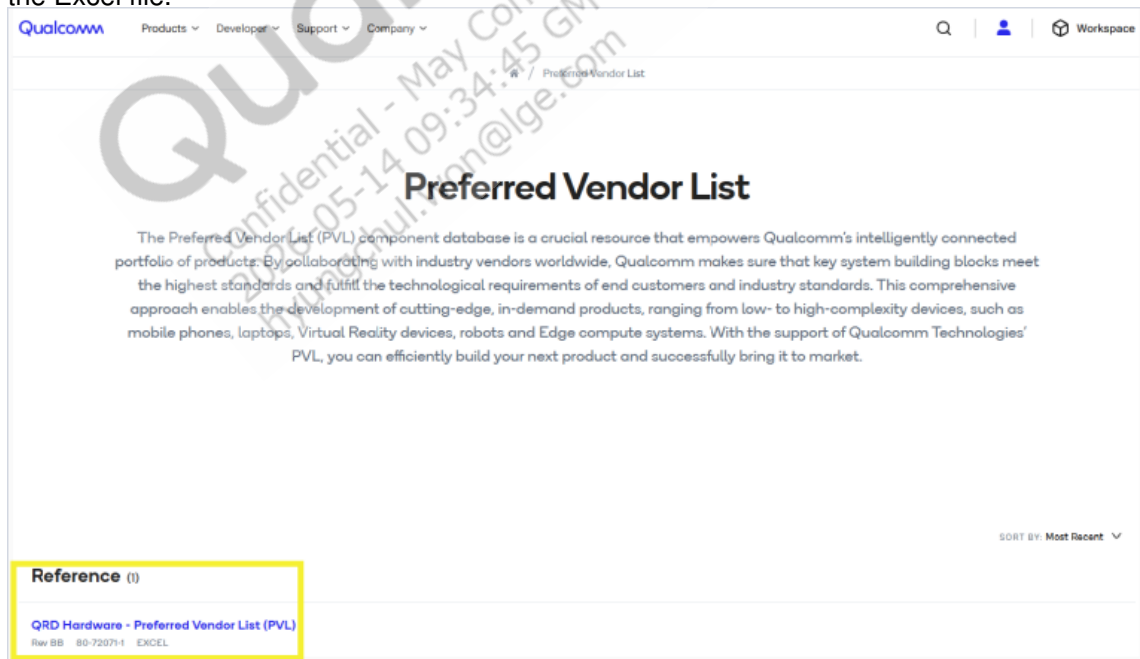
2 Customize and validate the DRAM components

This section explains how to use the preferred vendor list (PVL) to identify supported DRAM components and follow the required steps to select or validate custom DRAM.

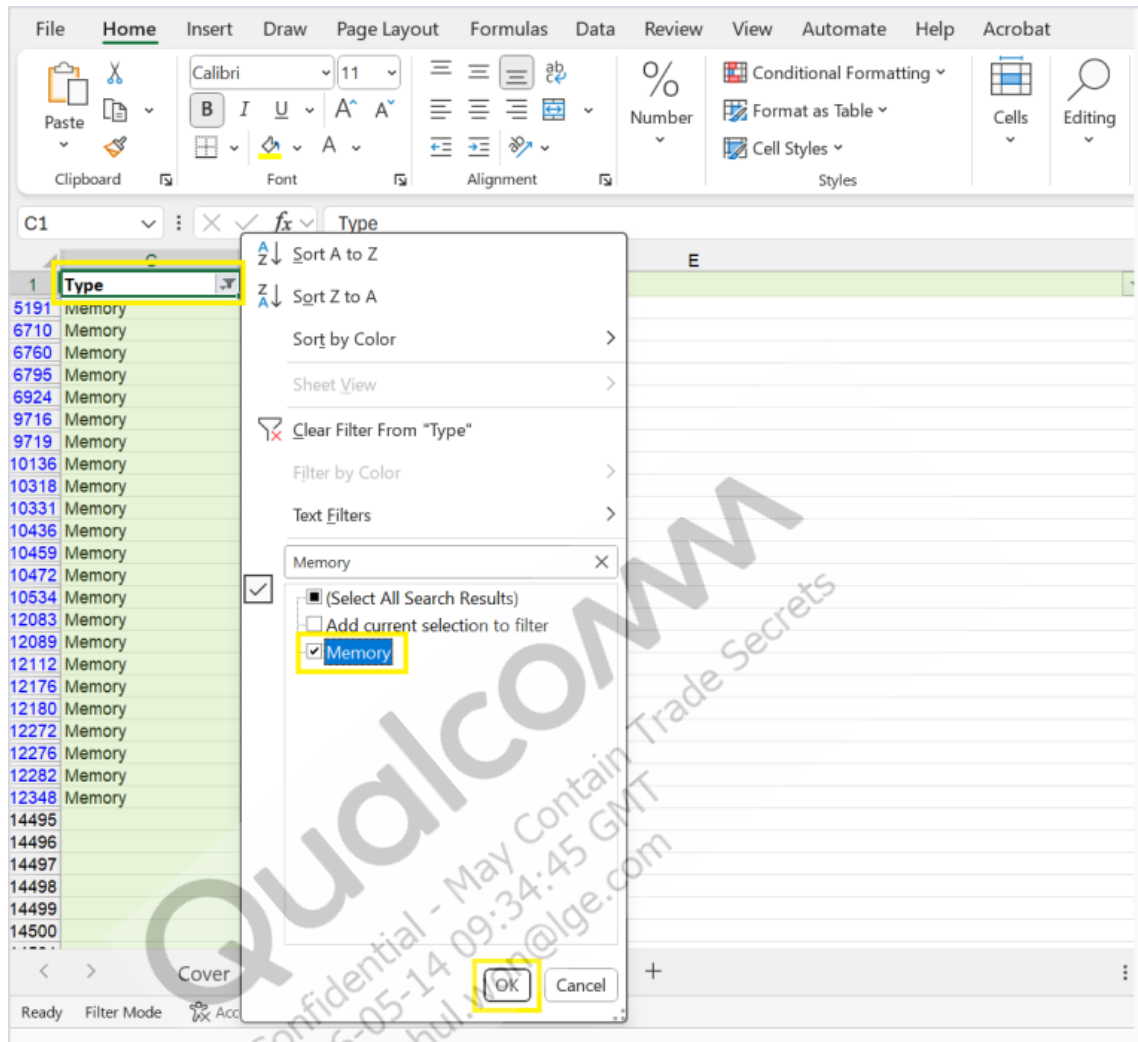
Qualcomm verifies and lists all DRAM components that work with its chip products in the PVL component database.

To find DRAM components in the PVL, do the following:

1. Open the [PVL](#) component database.
2. Sign in to your account.
3. In the **Reference** section, select **QRD Hardware - Preferred Vendor List (PVL)** to download the Excel file.



4. Open the downloaded Excel file and go to the **HWComponents** tab.
5. Apply an autofilter to the **Type** column, select **Memory**, and then select **OK** to confirm.



- Choose a specific <Chipset> in the **Chipset** column and then select **OK** to apply the autofilter option. For example, replace <Chipset> with SW6100.

Note:

The next release will include details about memory components supported for the SW6100 chipset PVL.

If you intend to use a different DRAM component that's not listed in the PVL for your product, contact [Qualcomm](#) for software support.

Qualcomm reviews your validation request, authenticates the component, and provides the appropriate software. After validation, use the provided software and tools such as the Qualcomm memory validation suite (QMVS). Use the QMVS tool to run a stress test and validate the DRAM.

For information about using the QMVS tool, see [Install and test the QMVS DDR validation tool](#).

3 Install and test the QMVS DDR validation tool

This section explains how to set up the Qualcomm Memory Validation Suite (QMVS) DDR tool and run automated tests to validate DRAM stability and performance. You'll find prerequisites, installation details, and instructions for executing test cases on the SW6100/SW6100P platform.

The QMVS DDR tool (version 1.0.0.0) provides automated validation for DDR memory.

Note:

- This release is intended only for DDR validation. Don't use it for FlashVal or storage validation.
- For FlashVal or storage-related testing, use the storage-specific QMVS package available under "Qualcomm Memory Validation Suite 2.0" in Qualcomm® Package Manager (QPM).

Prerequisites for using the QMVS tool

Before setting up the QMVS DDR validation tool, make sure the required software is installed.

Required software versions

Run the following commands to check the installed software versions:

Table: Required software versions

Software	Version	Command to check the installed version	Download
Node.js	10.24.1	<code>node --version</code>	https://nodejs.org/en/blog/release/v10.24.1
npm	6.14.12	<code>npm --version</code>	Included with Node.js
Python 3	3.8.4 (64-bit)	<code>python -V</code>	https://www.python.org/downloads/release/python-384/

Required Python packages

Install the following packages in your Python 3 environment:

```
pip install serial
pip install configparser
pip install pyserial
pip install psutil
pip install numpy
python -m pip install pywin32
```

Verify Python configuration

Run the following commands to confirm that your system is using Python 3.8.4:

```
python -V      # Should point to Python 3.8.4
python3 -V     # Should point to Python 3.8.4
py -3 -V      # Should point to Python 3.8.4
```

Environment configuration

Add the following directory to your system's PATH environment variable:

```
C:\Users\%USERPROFILE%\AppData\Roaming\npm
```

3.1 Download QMVS

To download the QMVS DDR package:

1. Open the [Qualcomm Package Manager \(QPM\)](#).
2. Sign in to your account.
3. Select **Tools** to find the *Qualcomm Memory Validation Suite 2.0*.
4. Download the Qualcomm Memory Validation Suite – DDR (DRAM) package on your target platform (Windows-x86).

The downloaded QMVS DDR tool package includes the following files:

Table: QMVS DDR package files

File	Description
<i>swsys-qmvs-ddr-vX.X.X.X.tgz</i>	QMVS DDR installation package
<i>QMVS_DDR_Usage_Guide_V1.0.0.pdf</i>	QMVS DDR user guide
<i>Stressapptest_Usage_Guide_V1.0.1.pdf</i>	Stressapptest user guide
<i>ReleaseNotes_and_IMP_Instruction.txt</i>	Release notes and instructions

3.2 Install QMVS

Run one of the following commands to install the QMVS DDR tool.

Option 1: Global installation

Use this option to install QMVS globally on your system:

```
npm install -g <path to swsys-qmvs-ddr-vX.X.X.X.tgz>
```

Option 2: Custom path installation

Use this option if you want to install QMVS to a specific folder:

```
npm install --prefix <path_where_to_install> <path_to_swsys-qmvs-ddr-vX.X.X.X>
```

Where:

```
path_where_to_install = C:\Users\%USERPROFILE%\AppData\Roaming\npm
```

3.3 Run QMVS validation tests

Use QMVS to run DDR validation test cases on the SW6100P platform.

Prerequisites to run QMVS validation tests

Before running the tests, ensure that:

- The QMVS DDR tool is installed. (see [Install QMVS](#))
- The target device is connected and accessible.

Supported test cases

QMVS supports the following test cases for SW6100P:

Table: Supported test cases

Test case	Description
QMESA	Qualcomm Memory Stress Application
STRESSAPPTTEST	Memory stress testing
DDR-CLK-SWITCH	DDR clock switching validation
CPU-CLK-SWITCH	CPU clock switching validation
REBOOT WITH RE-TRAIN	Reboot with DDR retraining

Test configuration files

Test configuration files are available in the following directory:

```
C:\Users\%USERPROFILE%\AppData\Roaming\npm\node_modules\swsys-qmvs\test_configs\SW6100_Configs\
```

The reboot with re-train configuration files are available in the following directory:

```
C:\Program Files (x86)\Qualcomm\QMVS_DDR\swsys-qmvs-ddr-v1.0.0.0.tgz\packages\test_configs
```

Run tests

Use the following command format to run any QMVS test:

```
test-suite -t <configuration file path>
```

Test command examples

Use the following examples to run QMVS test configurations on the SW6100P platform:

```
test-suite -t C:\Users\%USERPROFILE%\AppData\Roaming\npm\node_modules\swsys-qmvs\test_configs\SW6100_Configs\XXXX_Fixed_QMESA.json
test-suite -t C:\Users\%USERPROFILE%\AppData\Roaming\npm\node_modules\swsys-qmvs\test_configs\SW6100_Configs\XXXX_Floor_QMESA.json
test-suite -t C:\Users\%USERPROFILE%\AppData\Roaming\npm\node_modules\swsys-qmvs\test_configs\SW6100_Configs\XXXX_Fixed_Stressapptest.json
test-suite -t C:\Users\%USERPROFILE%\AppData\Roaming\npm\node_modules\swsys-qmvs\test_configs\SW6100_Configs\XXXX_Floor_Stressapptest.json
test-suite -t C:\Users\%USERPROFILE%\AppData\Roaming\npm\node_modules\swsys-qmvs\test_configs\Reboot_With_Retrain.json
```

Generate stressapptest binaries

The stressapptest binaries aren't included in this release due to compliance restrictions.

To generate the required binaries:

1. Open the “*Stressapptest_Usage_Guide_V1.0.1.pdf*” file available in the **Resources** folder.

```
<Path-to-installed QMVS-DDR Package>\QMVS_DDR\swsys-qmvs-ddr-v1.0.0.0.tgz\
packages\resources
```

2. Follow the naming conventions provided in the stressapptest user guide.
3. After you generate the binaries, copy them to the following directory:

```
<installed_path>\npm\node_modules\swsys-qmvs\node_modules\swsys-
stressapptest\bin
```

Note:

For more information, see the “*QMVS_DDR_Usage_Guide_V1.0.0.pdf*” included in the QMVS DDR tool package.

4 Troubleshoot DRAM issues

This section explains how to identify DRAM initialization and runtime issues by reviewing debug logs and memory usage data. It also provides examples to help you isolate and resolve common DRAM problems.

Qualcomm provides a sample log and an example command output to help with debugging.

Note:

For any DRAM-related issues, such as initialization failures, kernel panics, bit flips, corruptions, or if you are working with board designs featuring new DRAM, contact [Qualcomm](#) for software support.

4.1 DRAM initialization

The DRAM initializes sequentially during the XBL stage of bootup.

The following table lists the sequence of DRAM initialization during bootup as indicated in the serial universal asynchronous receiver/transmitter (UART) logs.

Table: DRAM initialization log entry and description

Log entry	Description
sbll_dds_init, Start	Starts DRAM initialization.
<LP5/LP5X> DDR detected	Detects a specific DRAM component.
Manufacturer ID = <XXXX>, Device Type = <XXXX>	Provides the manufacturing ID and device type of the DRAM.
Rank 0 size = <XXXX> MB, Rank 1 size = <XXXX> MB	Indicates the detection of the DRAM density.
Max enabled DDR Freq = <XXXX> MHz	Shows the maximum DRAM clock frequency.
DDR - Image Load, Start	Completes DRAM initialization and initiates image loading on the DRAM.
DDR Frequency, <XXXX> MHz	Sets the DRAM clock frequency to <XXXX> MHz before exiting the XBL. After the XBL exits, you can adjust the DRAM frequency to meet the requirement.

4.2 DRAM usage

To monitor real-time DRAM usage and available memory during runtime, run the following command:

```
adb root
adb shell
vienna64:/ cat /proc/meminfo
```

The following is a sample output:

```
vienna64:/ cat /proc/meminfo
MemTotal: <XXXX> kB
MemFree: <XXXX> kB
MemAvailable: <XXXX> kB
```

Note:

To update existing DRAM settings for debugging, see [QDUTT eCDT User Guide](#).

5 References

Find links to related resources, standards, and acronyms used in the guide.

5.1 Related documents

Title	Number
Qualcomm Technologies, Inc.	
QDUTT eCDT User Guide	80-NR058-9
Resources	
JEDEC standard	https://www.jedec.org/category/technology-focus-area/mobile-memory-lpddr-wide-io-memory-mcp
Preferred vendor list	https://www.qualcomm.com/preferred-vendor-list
Download Node.js	https://nodejs.org/en/blog/release/v10.24.1
Download Python 3	https://www.python.org/downloads/release/python-384/
Qualcomm Package Manager	https://qpm.qualcomm.com/#/main

5.2 Acronyms and terms

Acronym or term	Definition
ADB	Android debug bridge
DRAM	Dynamic random-access memory
DSF	DDR system firmware
LPDDR	Low-power double data rate
NPM	Node package manager
PoP	Package-on-package
PVL	Preferred vendor list
QMESA	Qualcomm Memory Stress Application
QMVS	Qualcomm Memory Validation Suite
QPM	Qualcomm Package Manager
XBL	eXtensible Boot Loader

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